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MySQL Implementation Essentials Bootcamp

MySQL Enterprise Monitor

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Program Agenda

- Introduction
- Oracle MySQL Enterprise
- Oracle Cloud Infrastructure overview
- Installation and Architecture
- Database Design
- Security
- Backup
- Replication
- High Availability
- **MySQL Enterprise Monitor**
- HeatWave

Monitor MySQL

(Community)

MySQL DBA Challenges

- “The database is slow. What needs tuning?”
- “What are my most expensive queries?”
- “Are indexes optimized?”
- “Is replication lag a problem?”
- “Did my last backup succeed?”
- “When will my disk fill up?”
- “When will I need more hardware to scale-out?”
- “Has my database schema changed?”
- “Are there security vulnerabilities that I need to be concerned about?”



MySQL DBA Checklist

1. Ensure your production databases are available ✓
2. Monitor MySQL performance throughout the day ✓
3. Verify that MySQL replication is working properly ✓
4. Confirm that backups have completed successfully ✓
5. Monitor disk space to ensure MySQL won't run out of space ✓
6. Regularly monitor and identify blocking issues ✓
7. Verify there have been no changes to database schema ✓
8. Check OS metrics for unusual events ✓
9. Check for security vulnerabilities ✓
10. Monitor and analyze memory usage ✓

Slow query log

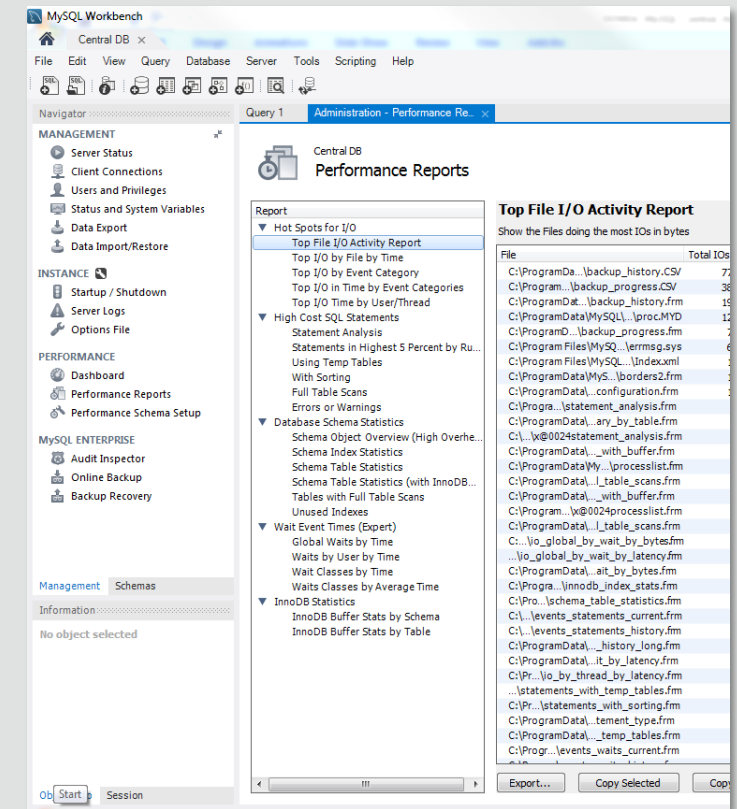
- The slow query log consists of SQL statements that take more than `long_query_time` seconds to execute and require at least `min_examined_row_limit` rows to be examined
 - By default, administrative statements are not logged, nor are queries that do not use indexes for lookups. This behavior can be changed using `log_slow_admin_statements` and `log_queries_not_using_indexes`
- The slow query log can be used to find queries that take a long time to execute and are therefore candidates for optimization.
- Examining a long slow query log can be a time-consuming task. To make this easier, you can use the `mysqldumpslow` command to process a slow query log file and summarize its contents
- Disabled by default

Performance Schema

- Provides runtime statistics to monitoring MySQL Server execution at a low level
- A new schema (database) called "performance_schema"
 - Introduced in MySQL 5.5
- Dedicated storage engine PERFORMANCE_SCHEMA
 - None persistent (restart of MySQL flush schema)
 - Nearly zero disk interaction
- Performance_schema is not replicated to replicas
- All recorded statistics are stored in fixed size ring buffers in memory

MySQL 8.0: SYS Schema

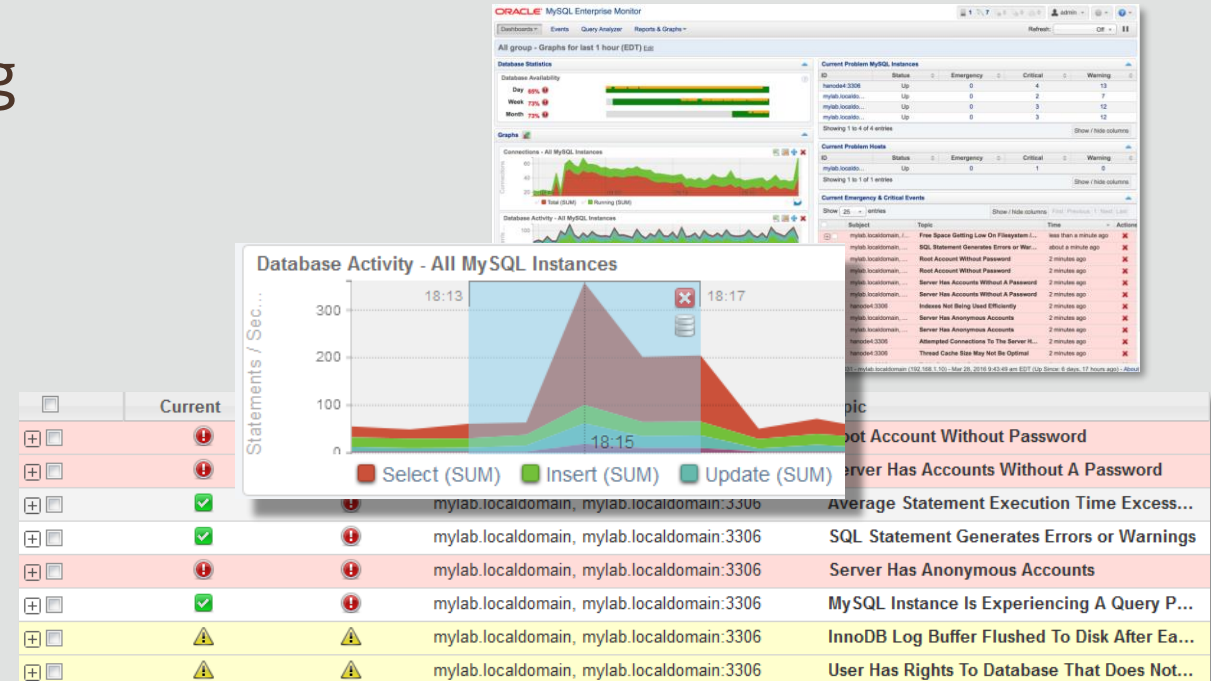
- Helper objects for DBAs, Developers and Operations staff
- Helps simplify DBA / Ops tasks
 - Monitor server health, user, host statistics
 - Spot, diagnose, and tune performance issues
- Easy to understand views with insights into
 - IO hot spots, Locking, Costly SQL statements
 - Schema, table and index statistics
- SYS is similar to
 - Oracle V\$ catalog views - primary interface for accessing information about system performance
 - Microsoft SQL DMVs (Dynamic Mgmt Views)



MySQL Enterprise Monitor

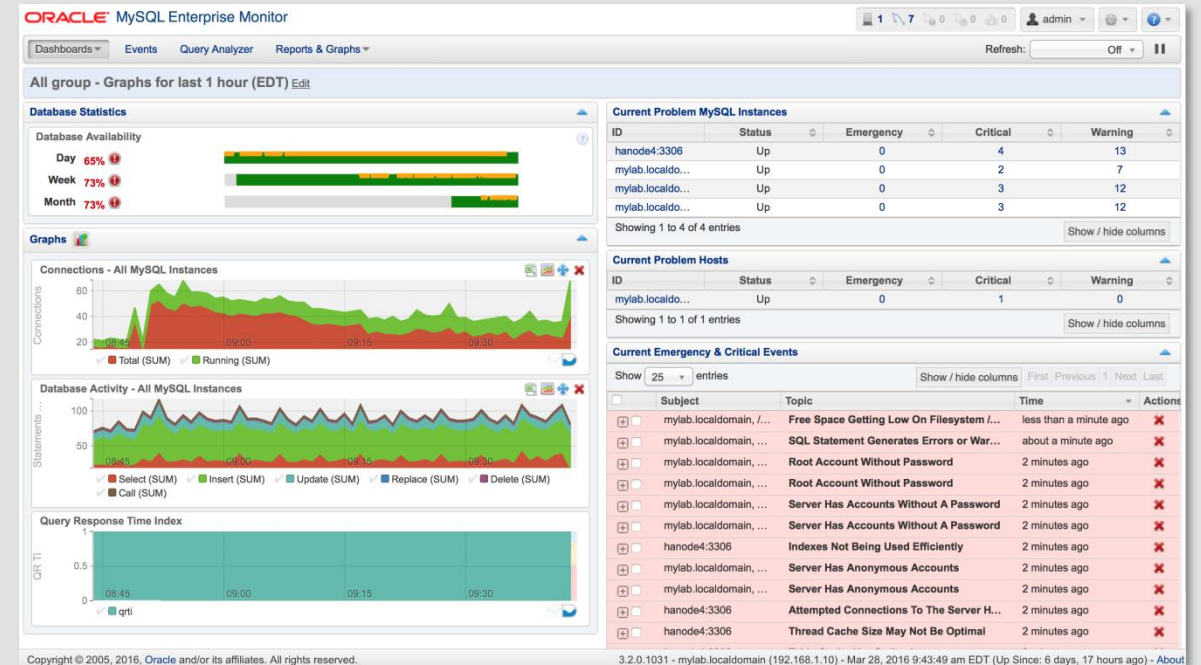
MySQL Enterprise Monitor

- Start monitoring (no management) MySQL in 10 minutes
- Real-time MySQL performance and availability monitoring
- Visually find & fix problem queries
- Disk monitoring for capacity planning
- Cloud friendly architecture
 - No agents required
 - Policy driven configuration
 - Easy integration with DevOps tools



MySQL Enterprise Dashboard

- SLA monitoring
- Real-time performance monitoring
- Alerts & notifications
- MySQL best practice advisors



"The MySQL Enterprise Monitor is an absolute must for any DBA who takes his work seriously."

Adrian Baumann
System Specialist
Federal Office of Information Technology & Telecommunications



Best Practice Advisors

- Enforce MySQL best practices
- 14 Advisor categories
- 250+ Advisors
- Threshold-based alerts
 - Exponential moving averages
 - Rate change detection
- Expert problem resolution advice

Advisors	
 Edit Selected Advisors	 Disable Selected Advisors
 Create Advisor	 Import/Export
Administration	Configured: 26 of 26
Agent	Configured: 3 of 3
Availability	Configured: 6 of 6
Backup	Configured: 5 of 5
Cluster	Configured: 10 of 10
Graphing	Configured: 87 of 87
Memory Usage	Configured: 6 of 6
Monitoring and Support Services	Configured: 5 of 5
Operating System	Configured: 5 of 5
Performance	Configured: 23 of 23
Query Analysis	Configured: 4 of 4
Replication	Configured: 13 of 13
Schema	Configured: 17 of 17
Security	Configured: 26 of 26

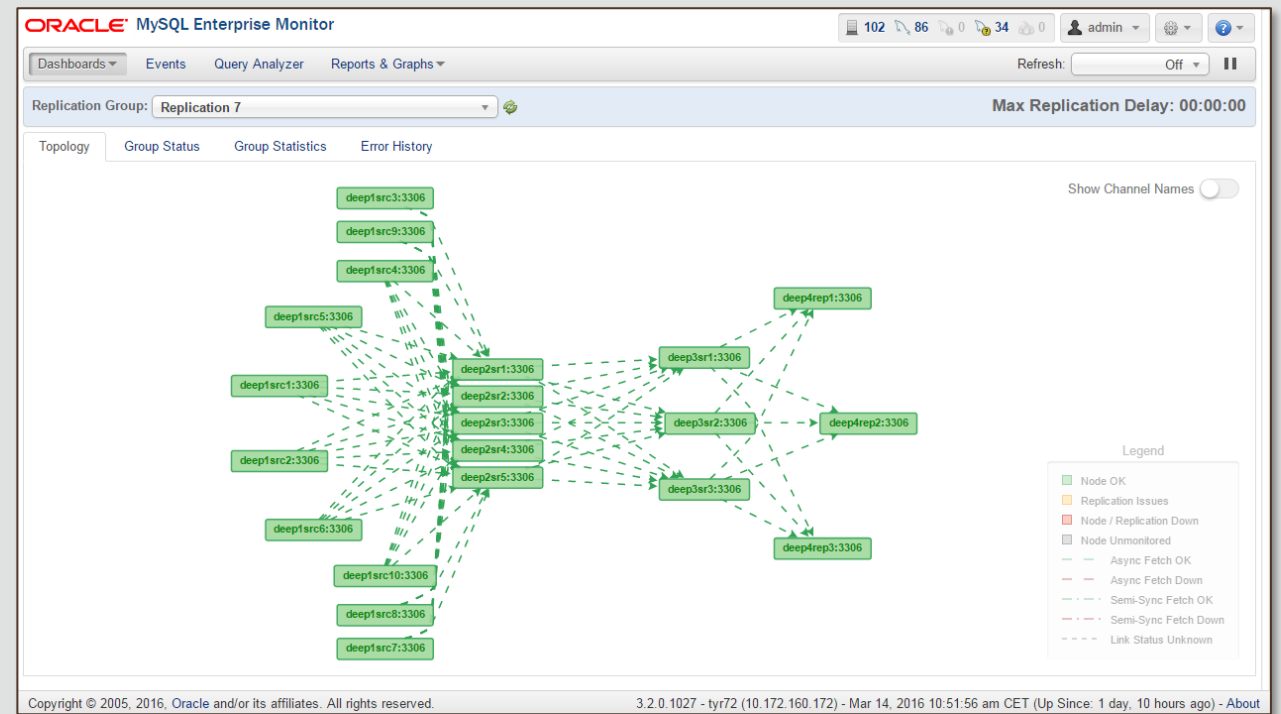
"I definitely recommend the MySQL Enterprise Monitor to DBAs who don't have a ton of MySQL experience. It makes monitoring MySQL security, performance and availability very easy to understand and to act on."

Sandi Barr
Sr. Software Engineer
Schneider Electric



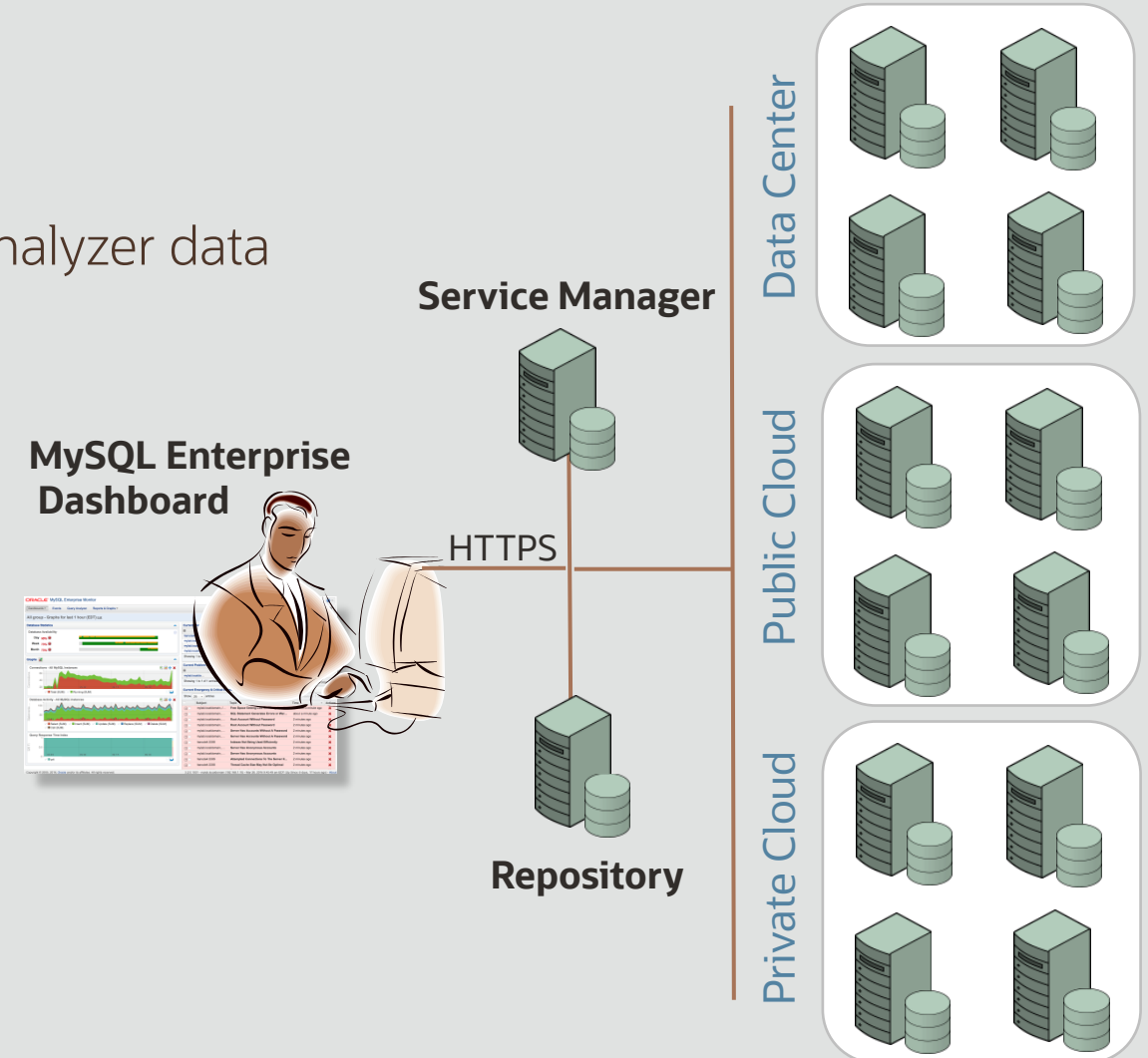
Replication Monitoring: Topology View

- Configuration of Replication Groups
- Replication Channel View
- Multiple Replication Topologies
 - Source/Replica
 - Multi-Source
 - Circular
 - Ring
 - Group Replication
 - NDB Cluster



Cloud Friendly Architecture

- **MySQL**
 - Performance Schema provides Query Analyzer data
 - Provides all MySQL related metrics
- **Service Manager**
 - Collects all MySQL related metrics
 - Collects all OS/Host related metrics
- **Repository**
 - Stores all historical data
- **Agent (optional)**
 - Only required for OS/Host metrics
 - Cache performance data



Installation & Configuration

- Install the Service Manager

```
shell> ./mysqlmonitor-8.X.X.XXXX-...-installer.bin
```

- Up and Running in 10 minutes

- Auto-discovery of MySQL servers
- Advisors are pre-configured
- Advisors are pre-scheduled

- Customization

- Centralized configuration dashboard
- Advisors
- Event handling



MySQL Enterprise Monitor Installation Options

- Text Mode (CLI based)

```
shell>./mysqlmonitor-8.0.14.1175-linux-x86_64-installer.bin --mode text
```

- Graphical mode (GUI based)

- Unattended Installation

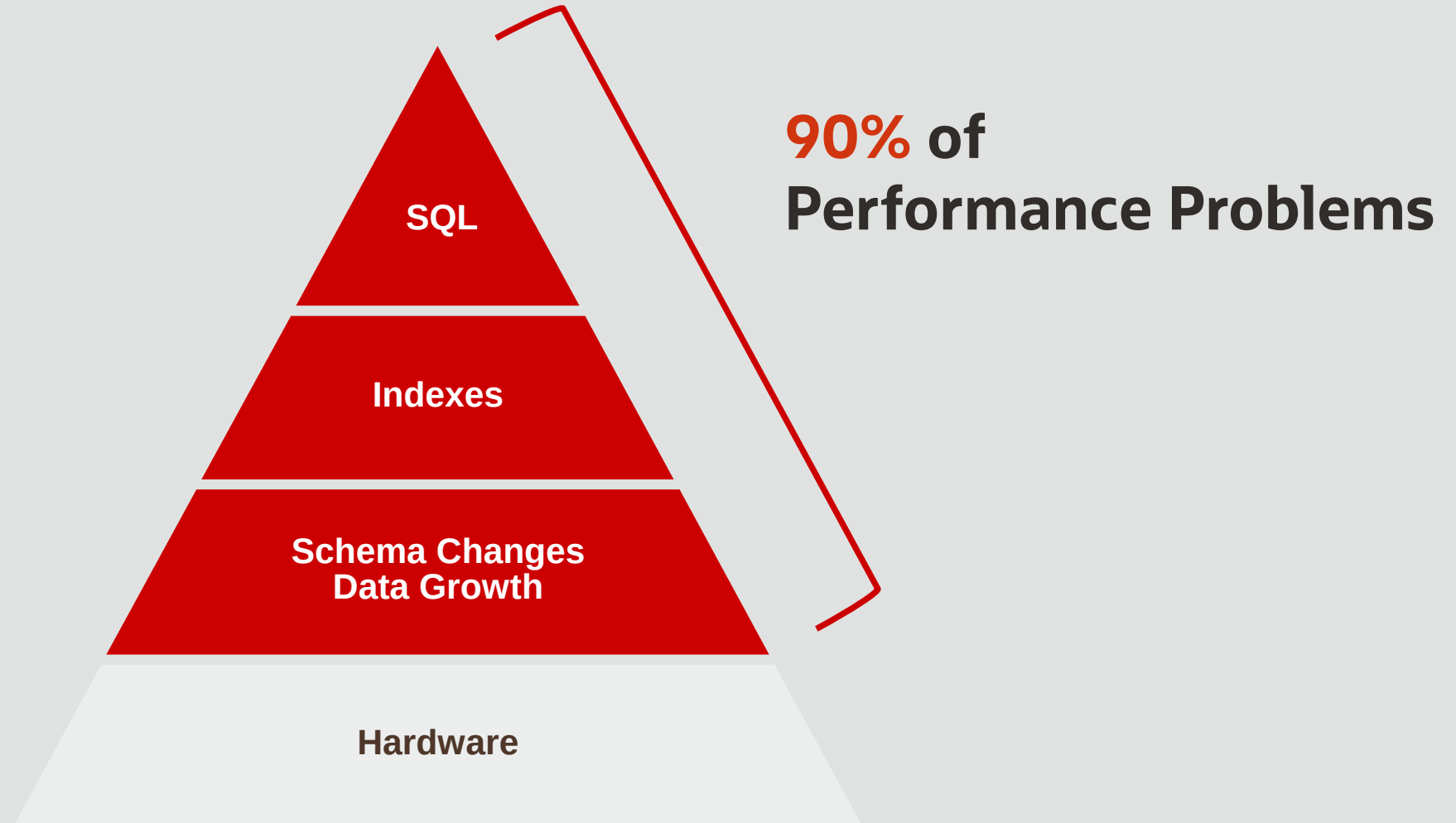
- specify `--unattended` (installation parameters: installation directory, user, password, and network options)
- For convenient scripting, you can save these options in a text file and run the installer using the `--optionfile` option

- Be careful with duplicated Server UUID

MySQL User Accounts for Monitor Agent

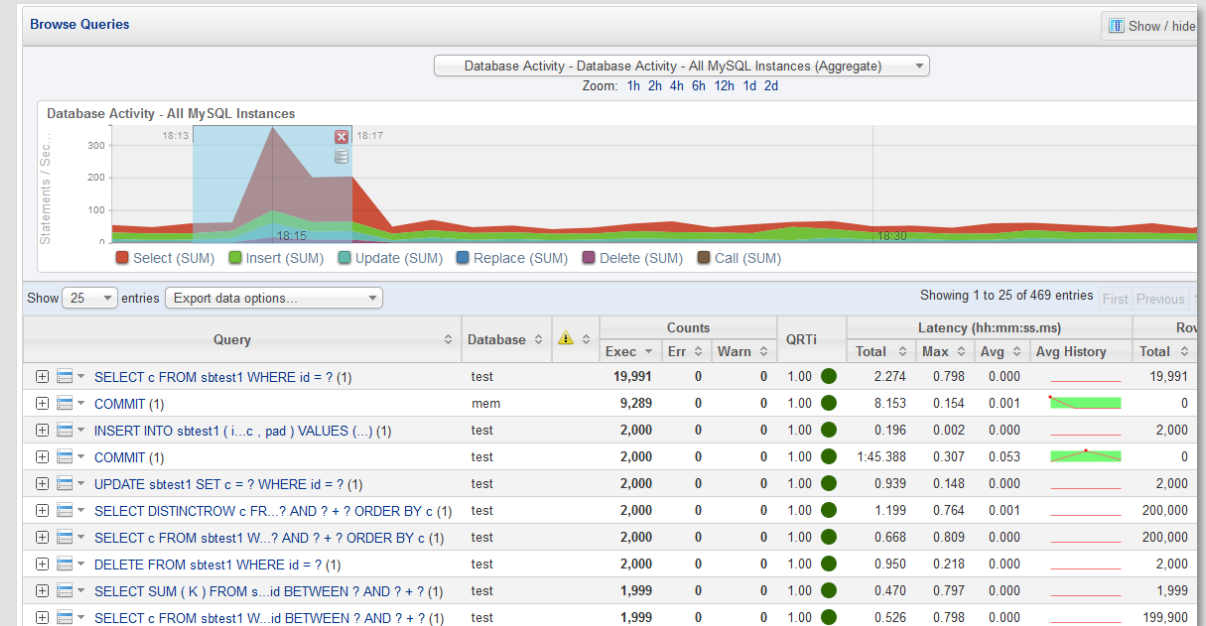
- Requires the **Admin** user
 - SUPER, CREATE, and INSERT privileges on the schema the inventory table
 - the inventory table stores unique identifiers for the MySQL instance and is created in the mysql schema by default
 - The SUPER privilege is required to temporarily switch off replication when creating and populating the inventory table, as well as running certain statements such as SHOW MASTER LOGS or SHOW ENGINE INNODB STATUS, depending on the version that is being monitored
- Optionally use **General** or **Limited** users, or both, depending on the system's security requirements

Source of Database Performance Problems



Enterprise Query Analyzer

- Real-time query performance
- Visual correlation graphs
- Find & fix expensive queries
- Detailed query statistics
- Query Response Time index (QRTi)



“With the MySQL Query Analyzer, we were able to identify and analyze problematic SQL code, and triple our database performance. More importantly, we were able to accomplish this in three days, rather than taking weeks.”

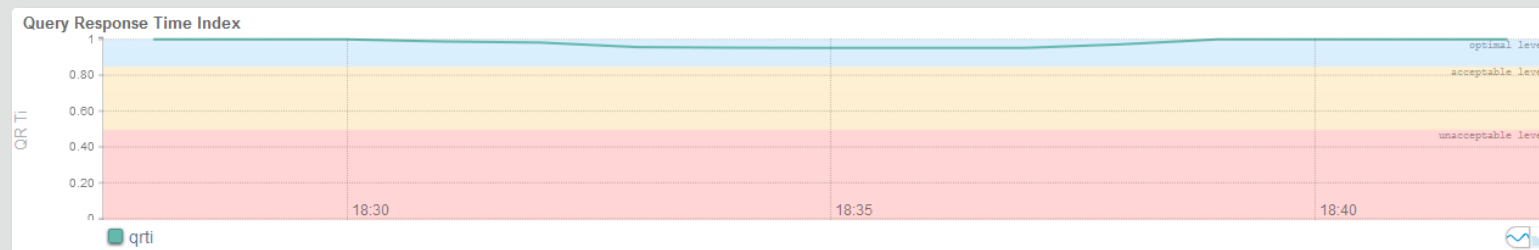
Keith Souhrada
Software Development Engineer
Big Fish Games



Query Response Time Index (QRTi)

- “Quality of Service” (QoS) measurement for each query
- QoS measurement for a server, group, or every instance
- Single metric for query performance
- Values (0..1)
- Low QRTi indicate low performance

Query	Database	Counts			QRTi
		Exec	Err	Warn	
INSERT INTO sbtest1 (i...c , pad) VALUES (...)	test	161,133	0	0	1.00
INSERT INTO sbtest1 (k...LUES (...) / * , ... */	test	20	0	0	0.50
SELECT SUM (K) FROM s...id BETWEEN ? AND ? + ?	test	161,156	0	0	1.00
SELECT DISTINCTROW c FR...? AND ? + ? ORDER BY c	test	161,160	0	0	1.00
CREATE INDEX k_1 ON sbtest1 (k)	test	1	0	0	0.00
SELECT c FROM sbtest1 W...? AND ? + ? ORDER BY c	test	161,157	0	0	1.00
BEGIN	test	161,139	0	0	1.00
CREATE TABLE sbtest1 (... = innodb MAX_ROWS = ?	test	1	0	0	0.50
COMMIT	test	161,091	0	0	0.69
DROP TABLE sbtest1	test	1	0	0	0.50



Providing Query Analyzer Data

- To supply queries information is used the Performance Schema statement digests with MySQL Server 5.6.14 and above
 - data are usually gathered directly
- **Verify the option**

```
UPDATE performance_schema.setup_consumers SET enabled = 'YES' WHERE name = 'statements_digest';
```
- **Alternatives for older versions**
 - Using a MySQL connector with a corresponding MySQL Enterprise Monitor Plugin that provides tracing and statistical information
 - Using the MySQL Enterprise Monitor Proxy and Aggregator

Performance Tuning of MySQL Enterprise Monitor

- **Tuning Tomcat**
 - Increase the amount of RAM available to the JVM installed with Tomcat
 - JvmMs/-Xms
 - JvmMx/-Xmx
 - If you change these settings, you must restart the MySQL Enterprise Service Manager.
- **Tuning InnoDB Memory**
 - Significant impact on performance, for both interaction with the Web UI, and overall resource requirements on the host.
 - increase the value of the innodb_buffer_pool_size variable

Oracle Enterprise Manager for MySQL

- Performance monitoring
- Availability monitoring
- Replication Monitoring
- InnoDB Monitoring
- Firewall Monitoring
- Audit Monitoring
- Reports
- Remote Monitoring
- Configuration monitoring
- All available metrics collected
- Allowing for custom threshold based incident reports



Hands-On Labs



8a - MySQL Enterprise Monitor - Install Service Manager

8b - MySQL Enterprise Monitor - Install Agent



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